Multidimensional Goodhart

A Response-Modeling Contract for Proxy Failure

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Abstract. Multidimensional Goodhart is not a theorem that more metrics help, hurt, or make residuals complex. The survived framework is conditional: proxy pressure licenses a calculation only after the response channel, action/search geometry, aggregation rule, hidden value or harm model, and falsifier are declared. The closed theorem set consists of pure-selection δ-reweighting bounds, a quadratic Stackelberg wedge, a convex score-deficit budget, an additive scorecard exchange-rate iff-condition, and a fixed finite deterministic adaptive-hardening boundary. The main negative result is equally important: score movement alone does not identify mechanism or welfare.

Motivation

Scalar Goodhart warnings compress two questions. First, the target is reduced to a proxy. Second, the residual between proxy and target is often treated as an unnamed loss. In multidimensional settings the residual is the object of interest: it has direction, support, exchange rates, active constraints, and evidence requirements. The question is therefore not whether proxy pressure can break a measure, but which response model makes a proposed calculation about hidden movement legal.

Write $G : S \to \mathbb{R}^m$ for target-relevant state, $P : S \to \mathbb{R}^k$ for proxy features, φ for the intended proxy relation, and $\varepsilon(s) = P(s) - \varphi(G(s))$ for residual proxy artifact. A claim must also say how policy exposure changes the population or behavior: selection changes weights $W_{\theta(u)}$ over fixed types, while intervention changes the fixed-type response kernel $K_{\theta(ds \mid u)}$.

Contract

The response-modeling contract is one paragraph: declare the type space U, baseline law and behavior, policy exposure, selection weights and/or response kernels, action/search geometry and stakes when interventions are claimed, proxy/target relation, aggregation rule, hidden value or harm model, evidence standard, and falsifier. Without those primitives, a Goodhart explanation is a retrospective label. With them, the framework says which of the following calculations can be imported and what it refuses to infer.

Closed Theorem Set

- **Selection bounds.** For pure selection with $\delta = \left\| d\frac{\mu_\theta}{d}\mu_0 - 1\right\|_{L^2(\mu_0)}$, hidden coordinates satisfy $\left|B_{H_i}\right| \leq \delta s_i$ and, after declaring the Euclidean coordinate norm, $\|B_H\|_2 \leq \delta \|s\|_2$. With a declared scalar value vector v, $|\Delta(v \cdot H)| \leq \delta\sqrt{v^T \Sigma_H v}$. The value metric is an input, not learned from score movement.
- **Intervention budgets.** In the one-dimensional quadratic threshold toy, gaming is worthwhile exactly within the wedge $\Delta = \sqrt{2\kappa V}$. More generally, with convex cost c and linear proxy gain $w \cdot a$, the private score-deficit cost is $m(d) = \sup_{\lambda \geq 0}[\lambda d - c^*(\lambda w)]$. This is not a welfare bound without hidden harm.
- **Scorecard exchange rate.** With additive score, separable quadratic costs, fixed deficit d, and hidden harm rates h_j, $H_{M(d)} = d\frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}$. Fixed-deficit per-agent harm is conserved across active measured sets iff $h_j = c w_j$ on those channels.
- **Adaptive hardening.** Under fixed finite M, fixed d, fixed V, fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone capacity hardening, gaming is feasible exactly when $S_{t(M)} = \sum_j \kappa_{j,t} w_j^2 \geq \frac{d^2}{2V}$ and stops exactly below that threshold.

Killed Claims

- Dimension count alone does not determine hidden harm.
- "More measured dimensions" has no sign without an aggregation rule and harm exchange rates.
- Baseline covariance is only a local velocity, not a finite-pressure selection primitive.
- Absolute continuity is not the causal intervention boundary.
- The selection/intervention split is not representation-free or marginally identifiable.
- Coordinate-explicit selection bounds are not welfare or coordinate-free claims until a value metric is declared.
- Convex affordability is not a hidden-harm bound.
- Generic minimum-complexity attraction is not a theorem.

Falsifier and Boundaries

The framework would be falsified by a domain where primitives are declared in advance, response channel and action traces distinguish nearby mechanisms, and the licensed calculation systematically predicts the wrong response shape or direction while a simpler score-only rule predicts it correctly. Narrow theorem failures would also falsify the relevant import: selection drift exceeding the δs_i envelope under its hypotheses; deterministic hardening violating the $S_{t(M)} < \frac{d^2}{2V}$ boundary; or additive fixed-deficit harm ignoring the $h_j = c w_j$ exchange-rate condition.

The framework refuses policy and welfare conclusions when primitives are unavailable. Examples include drifting type spaces, unobservable scorecard weights, changing measured sets, no defensible hidden value model, or applications where score movement is the only evidence.

Literature Relation and Open Problem

Goodhart, Campbell, Strathern, and Manheim–Garrabrant provide the genealogy and taxonomy. Lucas, strategic classification, performative prediction, multitask incentives, reward hacking, adaptive holdout work, and scalar tail-conditioned Goodhart provide formal analogues, but each supplies only some primitives. The framework's claim is reduction rather than unification: name the primitive a source contributes, and name what it omits.

The signature open problem is Q18, the residual-shape conjecture: under what declared response geometry, update rule, and predeclared complexity functional does repeated proxy repair drive hidden failure toward a predictable residual shape? The current answer is negative without those declarations.